Name: _____

TOPIC TEST 1

Chapter 6: Time series analysis

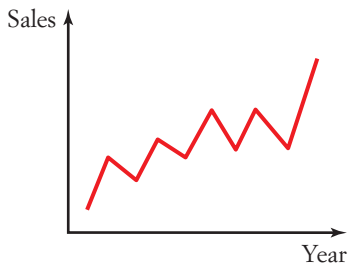
- Time allowed: 45 minutes
- CAS calculators may be used
- Part A: 8 multiple-choice questions (8 marks)
- Part B: 6 short-answer questions (34 marks)
- Part C: 2 analysis questions (20 marks)

Part A

Multiple-choice questions

8 multiple-choice questions
1 mark each: 8 marks
Circle the correct answer.

1 The time series graph shown below displays:



A Random fluctuations only

B Seasonal variation

C Cyclic variation

D A secular positive trend

E A secular negative trend

[1 mark]

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2 The time series graph shown below displays:



A Random fluctuations only

B Seasonal variation

C Cyclic variation

D A secular positive trend

E A secular negative trend

[1 mark]

3 The seasonal indices (SI) for sales of sunglasses by a factory outlet are shown in the table below.

Quarter	1	2	3	4
SI	1.23	0.97	0.66	1.14

The actual number of sunglasses sold in the fourth quarter was 163. The seasonally adjusted sales figure is closest to:

A 143

B 114

C 164

D 186

E 123

[1 mark]

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4 The seasonal indices (SI) of the sales from a retail outlet are shown in the table below.

Quarter	SI
1	0.9
2	1.4
3	1.3
4	0.4

For the first quarter of the year, the deseasonalised sales were $50 000. The actual sales figure for the first quarter was:

A $55 556

B $45 000

C $38 462

D $50 000

E $125 000

[1 mark]

5 For the points (1, 1), (2, 2.5), (4, 5.5), the linear regression line can be expressed as:

A $y = -1.4t + 6.25$

B $y = -1.5t + 6.25$

C $y = 1.5t + 0.5$

D $y = 1.5t - 0.5$

E $y = -3t + 8.5$

[1 mark]

6 For the following time series situation, which is most likely to apply?

Sales of gumboots in London

A Positive secular trend

B Negative secular trend

C Seasonal pattern

D Cyclic pattern

E Random pattern

[1 mark]

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7 A manufacturer of DVD players has determined the quarterly proportions shown below describing the shop's quarterly sales projected figures for 3 years.

Year	Q1	Q2	Q3	Q4
2014	0.70	0.86	1.08	1.36
2015	0.64	0.85	1.23	1.28
2016	0.53	0.91	1.45	1.11

The seasonal index for the first quarter is:

A 1.25

B 0.71

C 0.87

D 1.25

E 0.62

[1 mark]

8 A teacher wanted to analyse the number of absences in his classes. He listed the daily average absentee rates for Year 9 students for each term over a 2-year period and calculated these seasonal indices:

Year	Q1	Q2	Q3	Q4
Seasonalised index	0.60	0.95	1.08	1.37

If the equation describing the deseasonalised data is:

$$Absences = 1.2 \times Term + 27$$

the predicted actual number of absences in the first term of the third year is:

A 23

B 38

C 28

D 42

E 25

[1 mark]

Part B

Short-answer questions

6 short-answer questions
34 marks
Show your working where appropriate.

1 **a** Use the three-median method to smooth the following data set and plot the resulting graph.

t	1	2	3	4	5	6	7	8	9	10
y	3	5	6	10	9	16	12	16	17	20

b What type of trend is this?

[5 marks]

2 The following table shows the share price of a company during a volatile period in the stock market.

Day t	1	2	3	4	5	6	7	8	9	10
Price y ($)	3.23	5.55	6.36	10.1	9.1	16.2	12.1	16	17.25	10.1

a Use the three moving averages method to smooth the following data set. Plot the resulting graph.

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b What type of trend is this?

[5 marks]

3 Draw a time series plot and describe its features for the following data set.

t	1	2	3	4	5	6	7	8
y	3	5	6	10	1	4	5	16

[4 marks]

4 A holiday maker records the number of sightings, y, of rosellas in the Colac Otways that he observes each day, t, and constructs the following table.

t	1	2	3	4	5	6	7	8
y	3	5	6	1	1	4	5	1

a Draw a time series plot and describe its features.

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b Find the least squares regression line and add it to your time series plot.

[6 marks]

5 Staff in a small country hospital tabulated the number of births recorded over a period of 3 years as shown in the table below.

Year	Q1	Q2	Q3	Q4
2011	0	4	1	5
2012	1	5	1	4
2013	1	2	5	1

a Create a time series plot using the data given.

b Using technology, find the least squares regression line and add it to your time series plot.

c State the equation of the least squares regression line.

d Use the equation to predict the number of births in the small country hospital in the first quarter of 2014.

[8 marks]

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6 An ice-cream cone seller is trying to predict his sales figures for 2014. He is trying out a new liquorice flavour and has tabulated his sales figures of the liquorice flavour for the years 2008 to 2013:

Year t	2008	2009	2010	2011	2012	2013
Sales y ($1000s)	16	12	10	15	17	16

Using a four-point moving median with *centring*, smooth the data and use it to predict his sales for 2014.

[6 marks]

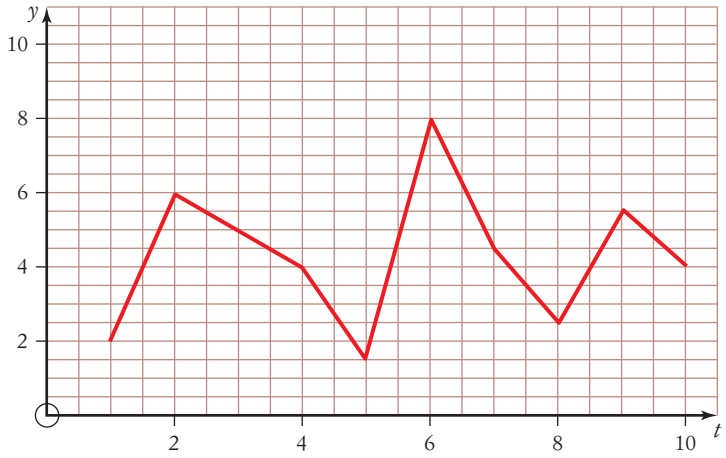
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Part C

Analysis questions

2 analysis questions
20 marks
Show your working where appropriate.

1 A teacher recorded the number of students absent per week for 10 days and the results are displayed in the graph below.



a Smooth the time series data using three-point median smoothing.

b Use the smoothed data to predict absences for day 11.

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c Smooth the time series data using a four-point moving average.

d Describe the trend shown.

[10 marks]

2 The quarterly sales figures (number of cars sold) recorded by a car sales yard for each of the years from 2010 to 2012 are shown in the table below.

Year	Sales figures quarter 1	Sales figures quarter 2	Sales figures quarter 3	Sales figures quarter 4
2010	5	7	9	3
2011	4	8	9	4
2012	5	9	10	5

a Calculate the seasonal index for each quarter.

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b Deseasonalise the data.

c Plot the original and the deseasonalised time series data.

[10 marks]

[Total marks: 62]

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Answers

Part A – Multiple-choice questions **[1 mark each]**

1 The time series data over the year shown is moving in an overall positive direction.

∴ **D**

2 The time series data shown is for a period of 3 months. If we were to divide the graph into three sections, one for each month, then we can see that there is a distinct pattern that repeats each month.

∴ **B**

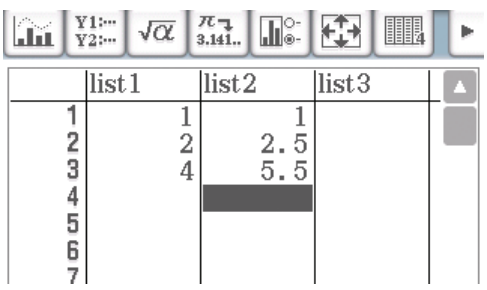
3 $$\text{Deseasonalised value} = \frac{\text{Actual value}}{\text{Seasonal index}}$$
$$= \frac{163}{1.14}$$
$$= 142.982\ldots$$
$$\approx 143$$

∴ **A**

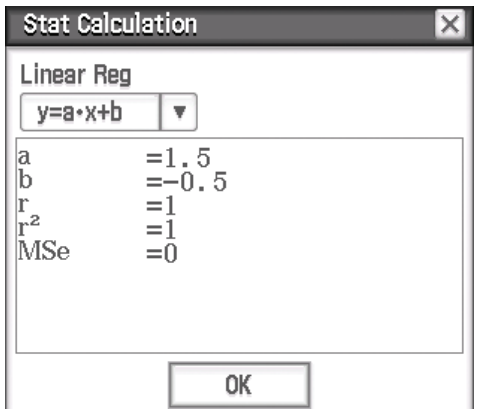
4 $$\text{Actual value} = \text{Seasonal index} \times \text{Deseasonalised value}$$
$$= 0.9 \times 50\,000$$
$$= 45\,000$$

∴ **B**

5 CAS required to plot the points and create the least squares regression line, as shown below.



	list1	list2	list3
1	1	1	
2	2	2.5	
3	4	5.5	
4			
5			
6			
7			



∴ **D**

6 A seasonal pattern is most likely to apply, as people are more likely to buy gumboots during rainy seasons.

∴ **C**

7 Seasonal index for first quarter
$$= \frac{0.70 + 0.64 + 0.53}{3}$$
$$= 0.62\dot{3}$$
$$\approx 0.62$$

∴ **E**

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8 $Absences = 1.2 \times Term + 27$

$= 1.2 \times 1 + 27$

$= 28.2$

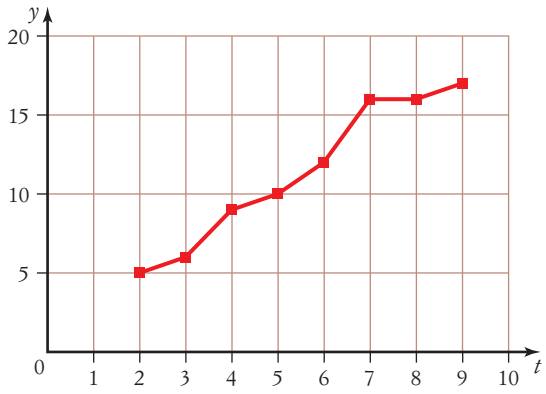
≈ 28

$\therefore$ C

Part B – Short-answer questions

1 a

t	1	2	3	4	5	6	7	8	9	10
y	3	5	6	10	9	16	12	16	17	20
		Med (3,5,6) = 5	Med (5,6,10) = 6	Med (6,10,9) = 9	Med (10,9,16) = 10	Med (9,16,12) = 12	Med (16,12,16) = 16	Med (12,16,17) = 16	Med (16,17,20) = 17	



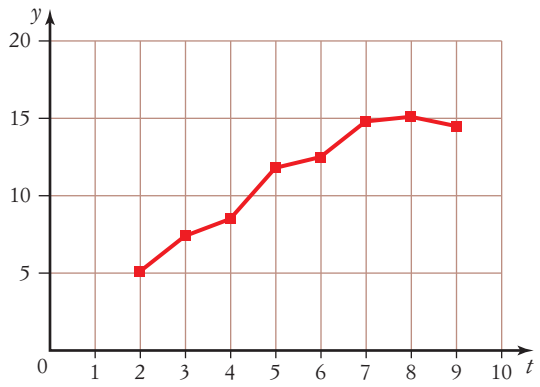
[4 marks]

b Positive secular trend **[1 mark]**

2

t	1	2	3	4	5	6	7	8	9	10
y ($)	3.2	5.6	6.4	10.1	9.1	16.2	12.1	16	17.3	10.1
		(3.2 + 5.6 + 6.4)/3	(5.6 + 6.4 + 10.1)/3	(6.4 + 10.1 + 9.1)/3	(10.1 + 9.1 + 16.2)/3	(9.1 + 16.2 + 12.1)/3	(16.2 + 12.1 + 16.0)/3	(12.1 + 16.0 + 17.3)/3	(16.0 + 17.3 + 10.1)/3	
		= 5.1	= 7.4	= 8.5	= 11.8	= 12.5	= 14.8	= 15.1	= 14.5	

a

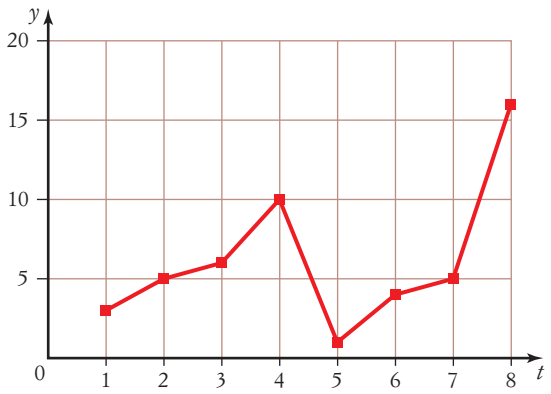


[4 marks]

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b Positive secular trend **[1 mark]**

3



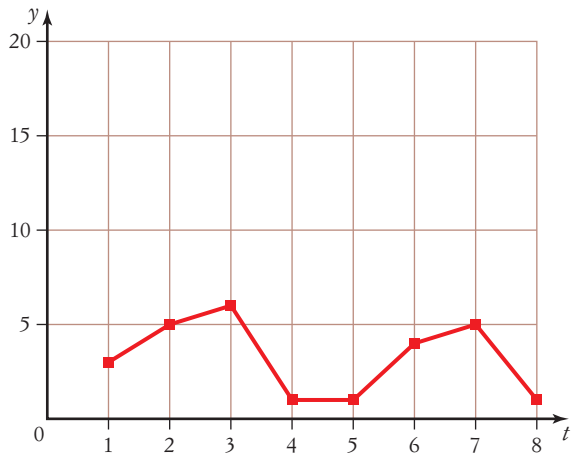
[3 marks]

Graph shows a cyclic trend peaking every fourth point. **[1 mark]**

4

t	1	2	3	4	5	6	7	8
y	3	5	6	1	1	4	5	1

a



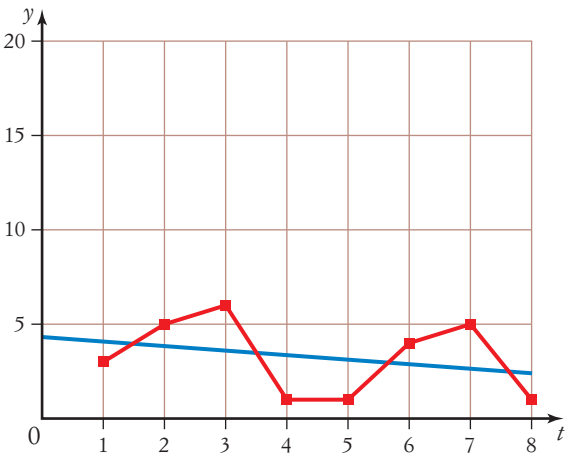
Plot has a lot of variation but, overall, rosella numbers are decreasing.

[3 marks]

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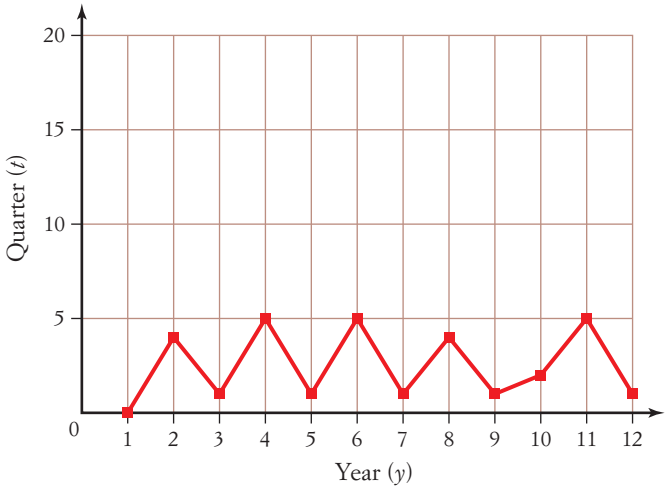
b Equation of least squares regression line is $y = -0.24t + 4.32$. **[1 mark]**

Addition to the time series plot of the least squares regression line is shown below.



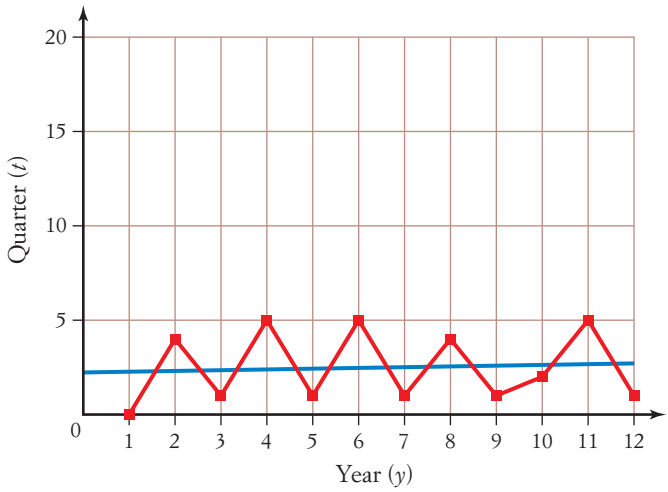
[2 marks]

5 a



[3 marks]

b



[2 marks]

c $y = 0.04t + 2.23$ **[1 mark]**

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d At $t = 13$, $y = 0.04 \times 13 + 2.23 = 2.75$

Round up to 3 births.

Predicted number of births for the first quarter of 2014 is three. **[2 marks]**

6

Year t/Sales y ($1000s)		Four-point moving median	Four-point moving median with centring
1	16		
2	12		
		Med (16,12,10,15) = 13.5	
3	10		13.5
		Med (12,10,15,17) = 13.5	
4	15		14.5
		Med (10,15,17,16) = 15.5	
5	17		
6	16		

[3 marks]

Trend line: $y = t + 12.5$ **[1 mark]**

Sales for 2014: $t = 7$

$$y = 7 + 12.5$$
$$= 19.5$$

$\therefore$ \$19 500 in sales predicted for 2014 **[2 marks]**

Part C – Analysis questions

1 a

Day t	Students absent y	Three-point median average
1	2	
2	6	Med (2,6,5) = 5
3	5	Med (6,5,5) = 5
4	5	Med (5,5,2) = 5
5	2	Med (5,2,8) = 5
6	8	Med (2,8,5) = 5
7	5	Med (8,5,3) = 5
8	3	Med (5,3,5) = 5
9	5	Med (3,5,4) = 4
10	4	

[3 marks]

b Trend line: $y = 5.33 - 0.08t$

When $t = 11$, $y = 4.5$. Round up to 5 absences.

Therefore, the predicted number of absences for day 11 is five.

[2 marks]

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c Smoothing the time series data using a four-point moving average gives the following:

Day t	Students absent y	Four-point moving average	Four-point moving average with centring
1	2		
2	6		
		$(2 + 6 + 5 + 5)/4 = 4.5$	
3	5		4.5
		$(6 + 5 + 5 + 2)/4 = 4.5$	
4	5		4.75
		$(5 + 5 + 2 + 8)/4 = 5$	
5	2		5
		$(5 + 2 + 8 + 5)/4 = 5$	
6	8		4.75
		$(2 + 8 + 5 + 3)/4 = 4.5$	
7	5		4.875
		$(8 + 5 + 3 + 5)/4 = 5.25$	
8	3		4.75
		$(5 + 3 + 5 + 4)/4 = 4.25$	
9	5		
10	4		

[4 marks]

d Trend is best described as cyclic. **[1 mark]**

2

Year	Sales figures quarter 1	Sales figures quarter 2	Sales figures quarter 3	Sales figures quarter 4	Yearly average
2010	5/6	7/6	9/6	3/6	$24/4 = 6$
2011	4/6.25	8/6.25	9/6.25	4/6.25	$25/4 = 6.25$
2012	5/7.25	9/7.25	10/7.25	5/7.25	$29/4 = 7.25$
SI	0.72	1.23	1.44	0.61	

a

SI	0.72	1.23	1.44	0.61	

[4 marks]

b

Year	Sales figures quarter 1	Sales figures quarter 2	Sales figures quarter 3	Sales figures quarter 4
2010	$5/0.72 = 6.94$	$7/1.23 = 5.69$	$9/1.44 = 6.25$	$3/0.61 = 4.92$
2011	$4/0.72 = 5.56$	$8/1.23 = 6.50$	$9/1.44 = 6.25$	$4/0.61 = 6.56$
2012	$5/0.72 = 6.94$	$9/1.23 = 7.32$	$10/1.44 = 6.94$	$5/0.61 = 8.20$

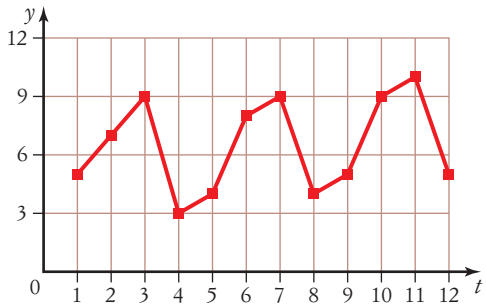
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Deseasonalised data

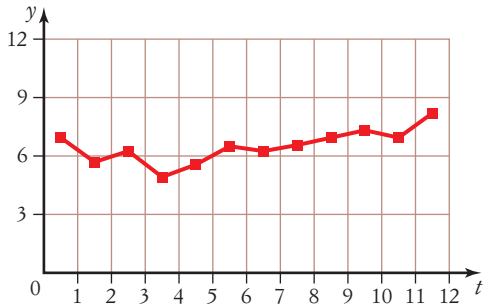
Year	Sales figures quarter 1	Sales figures quarter 2	Sales figures quarter 3	Sales figures quarter 4
2010	6.94	5.69	6.25	4.92
2011	5.56	6.50	6.25	6.56
2012	6.94	7.32	6.94	8.20

[4 marks]

c Original time series data



Deseasonalised time series data



[2 marks]

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